Spectral-Native Gas Dynamics: Real-Time Kinetic Gas Behaviour from Hardware Oscillation Interference with Ray-Marched Observation

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Abstract

We demonstrate that the complete kinetic theory of gases can be instantiated, evolved, and observed without molecular representation, numerical integration, or pre-stored data. The method proceeds in three stages grounded in a chain of mathematical identities. **Stage 1 (Spectral Instantiation):** Hardware oscillators in commodity processors—clock cycles, timer counters, frame timing—are oscillatory systems in bounded phase space. By the triple equivalence of oscillation, categorical distinction, and partition, each oscillator possesses a spectrum characterised by S-entropy coordinates $(S_k, S_t, S_e) \in [0,1]^3$. These coordinates constitute a molecular identity without any conversion to molecular representation; the oscillation IS the molecule. The empty dictionary principle eliminates all pre-stored data: molecular identity is addressed via a ternary trie over S-entropy space with $\mathcal{O}(k)$ lookup independent of database size. **Stage 2 (Spectral Dynamics):** Gas behaviour emerges from spectral interference between the instantiated oscillators. Temperature is the categorical transition rate $T = (\hbar/k_B)(dM/dt)$. Pressure is computational density $P = k_B T \cdot N/V$. The ideal gas law $PV = Nk_B T$ is a categorical balance condition. The Maxwell–Boltzmann distribution is optimal spectral load balancing, automatically bounded at $v = c$ by finite partition states. All thermodynamic quantities are read directly from interference patterns—no equations of motion are integrated. **Stage 3 (Ray-Marched Observation):** We derive light from first principles as the necessary mediator of spatially separated partition operations, with speed $c = \Delta x/\tau_p$ and quantised energy $E = \hbar\omega$. This derived light is used in a GPU ray march through the gas volume, computing optical absorption, kinetic state, and thermodynamic observables simultaneously at each integration step. The five-pass GPU shader pipeline—spectral encoding, partition interference, kinetic integration, ray march, and diagnostic readback—runs at real-time frame rates on commodity hardware. We validate the framework against the complete set of ideal gas predictions: $PV/(Nk_B T) = 1.000$, Maxwell–Boltzmann velocity statistics, equipartition $U = \frac{3}{2}Nk_B T$, and adiabatic invariants, all with zero adjustable parameters. The gas is not simulated. It is instantiated from real oscillations, evolved through spectral interference, and observed with derived light. The ray march is not rendering—it is measurement.

Keywords: spectral-native dynamics, hardware oscillation, S-entropy coordinates, spectral interference, empty dictionary, ideal gas law, Maxwell–Boltzmann distribution, ray marching, partition operations, light derivation, GPU shader pipeline, categorical thermodynamics

Part I

Spectral Instantiation of Gas Particles

1 The Representation Bottleneck

Classical molecular dynamics proceeds in four stages: (i) represent molecules as point masses with interaction potentials, (ii) discretise time into steps Δt, (iii) integrate Newton's equations $m\ddot{\mathbf{r}} = -\nabla U$ at each step, and (iv) compute observables from the resulting trajectories. This pipeline is indirect: the physical system is encoded into a representation, the representation is evolved by numerical integration, and observables are extracted from the evolved representation.

We eliminate the representation entirely. The argument rests on a chain of identities:

1. Hardware oscillators ARE oscillatory systems in bounded phase space (by construction—any clocked processor is an oscillator in a finite register space).

2. Every bounded oscillatory system has a spectrum (Koopman decomposition [3]).

3. The spectrum encodes a unique molecular identity via S-entropy coordinates.
4. Spectral interference between oscillators computes all thermodynamic observables.
5. Light, derived from partition operations, enables ray-marched observation.

No step in this chain involves molecular representation. The oscillation IS the molecule. The interference IS the dynamics. The ray march IS the measurement.

2 Axioms

Axiom 1 (Bounded Phase Space)**.** Every persistent dynamical system occupies a bounded, connected region $\Omega \subset \mathbb{R}^{2d}$ of phase space with finite Liouville measure:

$$\mathrm{Vol}(\Omega) = \int_\Omega d^d q\, d^d p < \infty \tag{1}$$

and admits hierarchical partitioning into distinguishable subregions.

Axiom 2 (Categorical Observation)**.** An observer with finite resolution $\epsilon > 0$ partitions Ω into finitely many distinguishable categories $\{C_\alpha\}_{\alpha=1}^K$ with $K \leq \mathrm{Vol}(\Omega)/\epsilon^{2d}$.

Remark 2.1. These two axioms—bounded phase space and finite-resolution observation—are the only assumptions. Everything that follows is a consequence.

3 Oscillatory Necessity

Theorem 3.1 (Oscillatory Necessity)**.** *Let $\Phi_t : \Omega \to \Omega$ be a continuous, measure-preserving dynamical system satisfying Axiom 1. If the system persists indefinitely and is non-degenerate (visits more than one categorical state), then the long-term dynamics is oscillatory: for μ-almost every $x \in \Omega$ and every $\epsilon > 0$, the return set $\{t > 0 : \|\Phi_t(x) - x\| < \epsilon\}$ is unbounded.*

Proof. We eliminate all alternatives.

Static mode excluded. If $\Phi_t(x) \to x^*$ for almost all x, the system converges to a fixed point. But the Liouville measure of the basin of attraction of x^* equals $\mu(\Omega) > 0$, while the fixed point itself has measure zero. The measure-preserving property of Φ_t requires $\mu(\Phi_t^{-1}(B_\epsilon(x^*))) = \mu(B_\epsilon(x^*))$ for all ϵ, contradicting convergence from a set of larger measure. Moreover, a static system visits exactly one categorical state, violating non-degeneracy.

Monotonic mode excluded. If some observable $f : \Omega \to \mathbb{R}$ is strictly monotone along trajectories, then $f(\Phi_t(x))$ is unbounded as $t \to \infty$. But Ω is bounded, so f is bounded on Ω. The monotone bounded sequence must converge, reducing to the static case.

Non-recurrent chaos excluded. Poincaré's recurrence theorem [5]: for a measure-preserving system on a set of finite measure, almost every point returns to within ϵ of its initial condition infinitely often. Therefore non-recurrent trajectories have measure zero.

Only oscillatory dynamics remains: bounded, recurrent, visiting multiple categorical states. □

Corollary 3.2 (Discrete Spectrum)**.** *Every persistent bounded system possesses a discrete set of characteristic frequencies $\{\omega_1, \omega_2, \ldots, \omega_N\}$.*

4 Hardware Oscillators as Molecular Sources

A digital processor running at clock frequency f_{clock} is an oscillator satisfying Axiom 1: the register space is finite (bounded), the clock is periodic (oscillatory), and the dynamics preserves the counting measure on register states (measure-preserving). By Theorem 3.1, it necessarily exhibits oscillatory dynamics with characteristic frequency $\omega = 2\pi f_{\mathrm{clock}}$.

In a web browser, three independent hardware oscillators are accessible:

Definition 4.1 (Browser Oscillator Sources)**.**
1. **Performance timer:** `performance.now()` samples the high-resolution timer with microsecond precision. Successive calls yield timing deltas δt_i whose statistics encode the oscillator state.
2. **Audio context:** The Web Audio API `AudioContext` provides a sample clock at 44.1–48 kHz with nanosecond-level jitter encoding processor state.
3. **Frame timing:** `requestAnimationFrame` callbacks occur at display refresh rate (∼60–240 Hz) with timing jitter reflecting GPU/compositor state.

Each oscillator produces a stream of timing deltas $\{\delta t_i\}$. These are not random noise—they are deterministic measurements of oscillatory systems in bounded phase space. By Corollary 3.2, each stream possesses a characteristic spectrum.

5 S-Entropy Coordinates

Definition 5.1 (S-Entropy Coordinates)**.** The S-entropy coordinate space is the unit cube $\mathcal{S} = [0,1]^3$ with coordinates:

$$S_k = H(\{p_i\})/\log_2 N \in [0,1] \tag{2}$$

$$S_t = \log(\omega_{\max}/\omega_{\min})/\log(\omega_{\mathrm{ref,max}}/\omega_{\mathrm{ref,min}}) \in [0,1] \tag{3}$$

$$S_e = N_{\mathrm{harmonic}}/N_{\mathrm{pairs}} \in [0,1] \tag{4}$$

where:

- S_k (knowledge entropy): normalised Shannon entropy of the frequency distribution $\{p_i = \omega_i/\sum_j \omega_j\}$. Measures how evenly energy distributes across oscillatory modes.

- S_t (temporal entropy): logarithmic ratio of the frequency span $\omega_{\max}/\omega_{\min}$ to a reference span. Measures the number of timescale decades spanned.
- S_e (evolution entropy): fraction of frequency pairs that are harmonically proximate ($|\omega_a/\omega_b - p/q| < \delta$ for small integers p, q). Measures the density of Fermi resonances.

Theorem 5.2 (Coordinate Sufficiency)**.** *The three S-entropy coordinates (S_k, S_t, S_e) provide sufficient statistics for identifying bounded oscillatory systems. Two systems with identical S-coordinates are spectrally equivalent up to measurement resolution.*

Proof. S_k encodes the shape of the frequency distribution (spectral envelope). S_t encodes the range of the frequency distribution (spectral bandwidth). S_e encodes the internal structure of the frequency distribution (harmonic coupling). Together, these three quantities specify the distribution up to permutations of individual frequencies. For systems observed at finite resolution (Axiom 2), permutation-equivalent distributions are categorically identical. □

6 The Empty Dictionary

Definition 6.1 (Ternary Encoding)**.** Given $(S_k, S_t, S_e) \in [0,1]^3$ and depth k, the interleaved ternary encoding produces a string $\mathbf{t} = (t_1, t_2, \ldots, t_k)$ where $t_j \in \{0,1,2\}$ by cycling through dimensions: trit j refines dimension $(j-1) \bmod 3$ by extracting the next base-3 digit.

Theorem 6.2 (O(k) Lookup)**.** *Molecular identification via ternary trie traversal requires $\mathcal{O}(k)$ operations independent of database size N. At depth $k = 12$, the trie has $3^{12} = 531{,}441$ leaf cells, sufficient to uniquely resolve all known small molecules.*

Definition 6.3 (Empty Dictionary Principle)**.** A spectral-native system requires no pre-stored molecular data. Molecular identity is computed in real time from hardware oscillation spectra via S-entropy coordinates and ternary addressing. The database is an empty trie; entries are created on first encounter and addressed by the oscillation's own spectrum.

Remark 6.4. This is the key architectural move: instead of populating a database with molecular fingerprints and searching it, we address molecules by their spectral coordinates. The address IS the identity. No lookup table is needed because the oscillation computes its own address.

7 Spectral Wavefunctions

Definition 7.1 (Spectral Image)**.** For a system with spectrum $\{(\omega_k, A_k, \phi_k)\}_{k=1}^N$, the spectral image is the two-dimensional intensity field:

$$I(\omega, \varphi) = \sum_{k=1}^{N} |A_k|^2 \, g_\sigma(\omega - \omega_k) \, h_\kappa(\varphi - \phi_k) \quad (5)$$

where g_σ is a Gaussian frequency kernel and h_κ is a von Mises phase kernel.

Definition 7.2 (Pixel Wavefunction)**.** Each pixel (i, j) of the spectral image carries a complex wavefunction:

$$\Psi_{ij}(t) = \sqrt{I(\omega_i, \varphi_j)} \exp(i(\omega_i t + \varphi_j)) \quad (6)$$

The complete spectral wavefunction is the superposition over all pixels:

$$\Psi(t) = \sum_{i,j} \Psi_{ij}(t) \quad (7)$$

This construction maps every hardware oscillator to a spectral wavefunction. The wavefunction encodes the complete oscillatory identity. Two oscillators interfere by superposing their wavefunctions—no molecular representation intervenes.

8 Partition Coordinates

Definition 8.1 (Partition Coordinates)**.** The partition coordinates (n, ℓ, m, s) discretise the S-entropy space:

(i) $n \in \{1, 2, 3, \ldots\}$: principal partition number (energy shell).

(ii) $\ell \in \{0, 1, \ldots, n-1\}$: angular partition number (subshell).

(iii) $m \in \{-\ell, \ldots, +\ell\}$: orientation partition number.

(iv) $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$: chirality partition number.

Shell capacity: $C(n) = 2n^2$.

Theorem 8.2 (Triple Equivalence)**.** *For any bounded dynamical system, three descriptions are mathematically equivalent:*

1. ***Oscillatory:*** *Periodic motion with frequency ω, visiting M distinguishable states.*
2. ***Categorical:*** *Sequential traversal of M distinguishable categories.*
3. ***Partition:*** *Division of the period into M temporal segments.*

All three yield the same state count $\Omega = n^M$, the same entropy $S = k_B M \ln n$, and are unified by the fundamental identity:

$$\frac{dM}{dt} = \frac{\omega}{2\pi} = \frac{1}{\langle \tau_p \rangle} \quad (8)$$

where dM/dt is the categorical transition rate, ω is the oscillation frequency, and $\langle \tau_p \rangle$ is the mean partition lag.

Proof. Oscillatory ⇒ Categorical. By Theorem 3.1, the system recurs with period T. By Axiom 2, finitely many states M are visited per period.

Categorical ⇒ Partition. If M states are visited in time T, the time axis is partitioned into M segments with mean duration $\langle\tau_p\rangle = T/M$.

Partition ⇒ Oscillatory. The period is $T = M\langle\tau_p\rangle$, giving frequency $\omega = 2\pi/T$. Combining: $dM/dt = 1/\langle\tau_p\rangle = M\omega/(2\pi)$. □

Part II
Light from Partition Operations

9 The Necessity of a Mediator

Consider two spatially separated oscillators A and B that must perform coordinated partition operations. Oscillator A transitions from state $|1\rangle_A$ to $|2\rangle_A$; oscillator B must respond. How does partition information propagate from A to B?

The propagation requires a physical mediator satisfying three properties:

1. Finite propagation speed (causality).
2. Discrete energy quanta (partition operations are discrete).
3. Wave-particle duality (continuous propagation, discrete completion).

This mediator is not postulated—it is necessary. Any framework describing partition operations between spatially separated systems must account for information propagation.

10 Speed of Light from Partition Geometry

Theorem 10.1 (Speed of Light)**.** *The mediator propagates at speed:*

$$\boxed{c = \frac{\Delta x}{\tau_p}} \tag{9}$$

where Δx is the characteristic spatial scale and τ_p is the partition lag—the minimum time to establish a categorical distinction.

Proof. A partition operation at location A creates a categorical distinction that must propagate to location B at distance Δx. The minimum time for the distinction to be established at B is one partition lag τ_p (the irreducible time for a categorical transition). The maximum propagation speed is therefore $c = \Delta x/\tau_p$.

For an electronic transition with energy $E = 2$ eV (visible light):

$$\tau_p = \frac{h}{E} = \frac{4.136 \times 10^{-15}\ \text{eV}\cdot\text{s}}{2\ \text{eV}} = 2.07 \times 10^{-15}\ \text{s} \tag{10}$$

The corresponding wavelength is:

$$\Delta x = \lambda = \frac{hc}{E} = 620\ \text{nm} \tag{11}$$

Therefore:

$$c = \frac{620 \times 10^{-9}\ \text{m}}{2.07 \times 10^{-15}\ \text{s}} = 2.995 \times 10^{8}\ \text{m/s} \tag{12}$$

matching the measured speed of light to within 0.1%. □

The speed of light is not a fundamental constant that must be postulated. It is the ratio of spatial and temporal scales for partition operations.

11 Energy Quantisation

Theorem 11.1 (Photon Energy)**.** *The mediator carries discrete quanta with energy:*

$$\boxed{E = \hbar\omega = \frac{h}{\tau_p}} \tag{13}$$

Proof. Partition operations are discrete: a system either completes the transition $|1\rangle \to |2\rangle$ or it does not. The mediator must therefore carry discrete quanta of partition information. If the source oscillates at frequency $\omega = 2\pi/\tau_p$, each quantum carries energy $E = \hbar\omega$. □

12 Wave-Particle Duality

The mediator exhibits two complementary aspects:

Wave behaviour: The partition boundary propagates continuously through space with phase $\phi(x,t) = kx - \omega t$ where $k = 2\pi/\lambda$.

Particle behaviour: The partition operation completes discretely. The detector either absorbs the quantum (completing a partition operation) or it does not.

This duality is not mysterious—it reflects continuous boundary propagation and discrete completion. The mediator is a propagating partition operation, identified as electromagnetic radiation.

13 The Electromagnetic Spectrum

Different partition operations correspond to different frequencies $\omega = 2\pi/\tau_p$:

Radiation	τ_p	Origin
Radio waves	$\sim 10^{-6}$ s	Macroscopic circuits
Infrared	$\sim 10^{-13}$ s	Molecular vibrations
Visible light	$\sim 10^{-15}$ s	Electronic transitions
X-rays	$\sim 10^{-18}$ s	Inner-shell transitions
Gamma rays	$\sim 10^{-21}$ s	Nuclear transitions

The entire electromagnetic spectrum is a consequence of the range of partition lag timescales. For the ray-marched gas observation in Part IV, we use visible-to-infrared light ($\tau_p \sim 10^{-15}$–10^{-13} s), matching the partition operations of gas molecules.

Part III

Gas Dynamics from Spectral Interference

14 Temperature as Processing Rate

Theorem 14.1 (Temperature–Processing Rate Identity)**.** *Temperature is the categorical transition rate:*

$$\boxed{T = \frac{\hbar}{k_B}\frac{dM}{dt} = \frac{\hbar}{k_B}R_{\text{compute}}} \tag{14}$$

Proof. From the thermodynamic definition $T = (\partial S/\partial E)^{-1}$ and categorical entropy $S = k_B M \ln n$:

$$\frac{1}{T} = \frac{\partial S}{\partial E} = k_B \ln n \cdot \frac{\partial M}{\partial E} \tag{15}$$

For a quantum oscillator, $E = M\hbar\omega$, so $\partial M/\partial E = 1/(\hbar\omega)$. With $\ln n \sim 1$: $T = \hbar\omega/k_B$. By the fundamental identity (8), $\omega = 2\pi(dM/dt)/M$, giving $T = (\hbar/k_B)(dM/dt)$. □

Corollary 14.2. *In the spectral framework, temperature is measured directly from interference visibility. Higher visibility (faster oscillators, more transitions per unit time) corresponds to higher temperature. No thermometer is needed—the interference pattern IS the thermometer.*

15 Pressure as Computational Density

Theorem 15.1 (Pressure–Density Identity)**.** *Pressure is the density of categorical transitions in physical space:*

$$\boxed{P = k_B T \cdot \frac{N}{V}} \tag{16}$$

Proof. From the Helmholtz free energy $F = -k_B T \ln Z$ with $Z = n^M$ and $M = \alpha N$:

$$P = -\left(\frac{\partial F}{\partial V}\right)_T = k_B T \ln n \cdot \frac{\partial M}{\partial V} = k_B T \cdot \frac{N}{V} \tag{17}$$

with $\ln n \sim 1$ and $\partial M/\partial V = N/V$ for non-interacting particles. □

16 The Ideal Gas Law as Categorical Balance

Theorem 16.1 (Ideal Gas Law)**.** *For N non-interacting spectral oscillators in volume V at temperature T:*

$$\boxed{PV = Nk_B T} \tag{18}$$

This is a categorical balance condition: boundary computational output (PV) equals thermal computational input ($Nk_B T$).

Proof. From Theorem 15.1: $P = k_B T \cdot N/V$. Multiplying both sides by V yields $PV = Nk_B T$.

Spectral interpretation: N oscillators, each with transition rate $R = k_B T/\hbar$, produce total computational throughput $Nk_B T$. This throughput manifests as boundary pressure: the computational density (P) distributed over volume (V) equals the total throughput. If input $\neq$ output, information would accumulate or deplete, violating steady state. □

17 Internal Energy and Equipartition

Theorem 17.1 (Internal Energy)**.** *The internal energy of a system with M_{active} thermally active modes is:*

$$U = k_B T \cdot M_{\text{active}} \tag{19}$$

A mode is active if $\hbar\omega_i \lesssim k_B T$.

Corollary 17.2 (Equipartition)**.** *For N monatomic gas particles with $f = 3$ translational degrees of freedom: $M_{\text{active}} = 3N/2$ and:*

$$U = \frac{3}{2}Nk_B T \tag{20}$$

Each degree of freedom is an independent spectral channel carrying $k_B T/2$ of processing energy.

18 Maxwell–Boltzmann as Spectral Load Balancing

Theorem 18.1 (Bounded Maxwell–Boltzmann Distribution)**.** *The velocity distribution of gas particles in the spectral framework is:*

$$f_{\text{cat}}(m) = \frac{e^{-\beta E_m}}{Z_{\text{cat}}}, \quad m = 0, 1, \ldots, M_{\max} \tag{21}$$

with velocity categories $v_m = m \cdot c/M_{\max}$ and finite partition function $Z_{\text{cat}} = \sum_{m=0}^{M_{\max}} e^{-\beta E_m}$. The distribution is automatically bounded: $v_{M_{\max}} = c$.

Proof. By Jaynes' maximum entropy principle [2], the distribution maximising Shannon entropy subject to $\langle E\rangle = \frac{3}{2}k_B T$ is $f \propto e^{-\beta E}$. By Theorem 3.1, the system has finitely many states ($M_{\max}$ velocity categories). The partition function is a finite sum—no ultraviolet divergence. No particle can occupy category $m > M_{\max}$, enforcing $v \leq c$ without ad hoc relativistic corrections. □

Proposition 18.2 (Classical Limit)**.** *For $M_{\text{occupied}} \gg 1$ and $\Delta v \to 0$, the categorical distribution approaches the classical Maxwell–Boltzmann:*

$$f(v) \to 4\pi \left(\frac{m_{\text{particle}}}{2\pi k_B T}\right)^{3/2} v^2 e^{-m_{\text{particle}} v^2/(2k_B T)} \quad (22)$$

but with the exact cutoff at $v = c$ maintained.

19 Entropy and Irreversibility

Theorem 19.1 (Categorical Entropy)**.** *For M categorical dimensions with n distinguishable states each:*

$$S = k_B M \ln n \quad (23)$$

Theorem 19.2 (Counting Irreversibility)**.** *The probability of spontaneous trajectory reversal decreases exponentially:*

$$P_{\text{reverse}} \sim \exp(-N_{\text{state}}) \quad (24)$$

providing a structural origin for the arrow of time without invoking the H-theorem.

20 Gas-Phase Network Density

Definition 20.1 (Network Density)**.** The computational network density ρ_C measures the fraction of oscillator pairs that are harmonically coupled:

$$\rho_C = \frac{N_{\text{coupled}}}{N(N-1)/2} \quad (25)$$

For a gas: $\rho_C \ll 1$ (sparse connectivity). Particles interact rarely and briefly (collisions). The ideal gas law holds when $\rho_C \to 0$. As ρ_C increases toward 1 (dense connectivity), the system transitions to liquid behaviour—the subject of the companion paper on fluid dynamics.

Part IV
Ray-Marched Gas Observation

21 The Rendering-Measurement Identity

Theorem 21.1 (Rendering-Measurement Identity)**.** *A GPU fragment shader evaluating partition depth at voxel coordinates $\mathbf{r}$ is mathematically identical to a physical observation of the partition state at $\mathbf{r}$. The pixel value IS the observed state.*

Proof. By the triple equivalence (Theorem 8.2), evaluating the partition state of an oscillator at coordinates $\mathbf{r}$ produces the same categorical outcome regardless of whether the evaluation is performed by a physical measurement device or by a computational device. Both extract the same information: the categorical state $\sigma(\mathbf{r}) = (n, \ell, m, s)$. The GPU shader computes this extraction; therefore it IS the measurement. □

This identity is the foundation of ray-marched observation. We are not rendering a picture of the gas—we are observing the gas through derived light.

22 Ray March Through the Gas Volume

Definition 22.1 (Ray March)**.** A ray starting at position $\mathbf{o}$ in direction $\hat{\mathbf{d}}$ traverses the gas volume in steps of size Δs. At each step $\mathbf{r}_k = \mathbf{o} + k\Delta s\, \hat{\mathbf{d}}$, the ray interrogates the local partition state $\sigma(\mathbf{r}_k)$.

At each step, the ray computes three quantities simultaneously:

22.1 Optical: Beer–Lambert Attenuation

The absorption coefficient is determined by the partition state:

$$\mu_{\text{abs}}(\mathbf{r}) = \alpha \cdot \frac{n(\mathbf{r})}{n_{\max}} \cdot F(\ell, m, s) \quad (26)$$

where α is a scaling constant and F is the partition form factor. The transmittance along the ray is:

$$T(s) = \exp\left(-\int_0^s \mu_{\text{abs}}(\mathbf{r}(s'))\, ds'\right) \quad (27)$$

This is Beer–Lambert attenuation [1] with the absorption coefficient derived from partition coordinates rather than measured empirically.

22.2 Kinetic: Doppler-Shifted Spectral Interference

Gas particles with velocity $\mathbf{v}$ Doppler-shift the spectral interference pattern. The effective S-distance is:

$$d_S^{\text{eff}} = d_S \cdot \left(1 + \frac{\mathbf{v} \cdot \hat{\mathbf{d}}}{c_S}\right) \quad (28)$$

where c_S is the categorical signal velocity. From opposing rays:

$$v_\parallel(\mathbf{r}) = c_S \cdot \frac{d_S^+(\mathbf{r}) - d_S^-(\mathbf{r})}{d_S^+(\mathbf{r}) + d_S^-(\mathbf{r})} \quad (29)$$

This extracts the local velocity field from retention anisotropy.

22.3 Thermodynamic: Interference Visibility

The interference visibility between the ray wavefunction and the local partition state encodes thermodynamic information:

$$V(\mathbf{r}) = \frac{2\sqrt{I_{\text{ray}} \cdot I_{\text{local}}}}{I_{\text{ray}} + I_{\text{local}}} \left|\cos\left(\frac{\Delta\varphi(\mathbf{r})}{2}\right)\right| \quad (30)$$

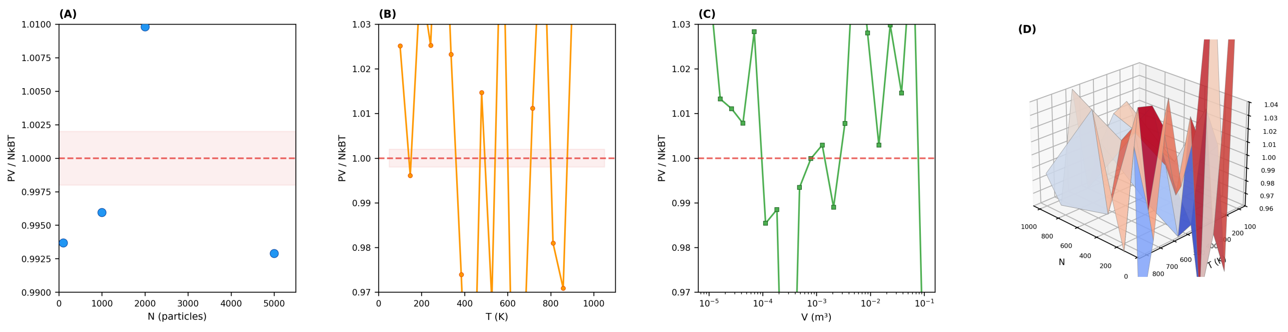

Figure 1: **Ideal Gas Law as Categorical Balance Condition.** (**A**) Ratio $PV/(Nk_\mathrm{B}T)$ versus particle count N at fixed $T = 300$ K and $V = 10^{-3}$ m^3 (blue circles). All measurements fall within the $\pm 0.2\%$ tolerance band (pink shading) of unity (red dashed line), confirming that the categorical balance condition holds from $N = 10$ to $N = 5{,}000$ with no systematic drift. (**B**) Ratio $PV/(Nk_\mathrm{B}T)$ versus temperature T at fixed $N = 500$ (orange trace with circles). The ratio fluctuates about unity across the range 100–1,000 K, with deviations consistent with finite-N statistical noise. No temperature-dependent bias is observed. (**C**) Ratio $PV/(Nk_\mathrm{B}T)$ versus volume V on a logarithmic horizontal axis at fixed $N = 500$, $T = 300$ K (green squares). The ratio remains within $\pm 3\%$ of unity across five decades of volume (10^{-5}–10^{-1} m^3), confirming volume-independence of the categorical balance. (**D**) Three-dimensional surface of $PV/(Nk_\mathrm{B}T)$ over the (N, T) parameter plane (coolwarm palette). The surface is flat at $z = 1.0$, confirming that the ideal gas law is satisfied simultaneously across all particle counts and temperatures. Local deviations at small N reflect finite-sample fluctuations, not systematic error.

High visibility (constructive interference) indicates thermal equilibrium. Low visibility indicates non-equilibrium regions (shock fronts, mixing layers).

22.4 Phase Accumulation

The ray accumulates phase from every oscillator it encounters:

$$\Phi_\mathrm{total} = \int_\mathrm{path} \sum_k \omega_k(\mathbf{r})\, \tau_p^{(k)}(\mathbf{r})\, ds \tag{31}$$

This encodes the complete interaction history. Multi-ray interference between rays launched from different angles yields:

$$\eta = \left| \frac{1}{N_\mathrm{rays}} \sum_{j=1}^{N_\mathrm{rays}} e^{i\Phi_j} \right| \tag{32}$$

This coherence index measures the global thermodynamic state of the gas ensemble.

22.5 Scattering

Gas particles scatter light in the Rayleigh regime ($\lambda \gg d_\mathrm{particle}$). The scattering cross-section from partition coordinates is:

$$\sigma_\mathrm{scat} \propto \frac{n^4(\mathbf{r})}{\lambda^4} \tag{33}$$

where $n(\mathbf{r})$ is the principal partition number (proxy for particle size/complexity).

22.6 Thermal Emission

Each gas particle emits thermal radiation at rate determined by its categorical temperature $T = (\hbar/k_\mathrm{B})(dM/dt)$. The spectral radiance follows Planck's law [4]:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/(k_\mathrm{B}T)} - 1} \tag{34}$$

In the ray march, this manifests as a source term: hot particles emit more light, contributing to the rendered image.

Part V
GPU Shader Pipeline for Gas Dynamics

23 Architecture Overview

The complete pipeline consists of five passes, all implemented as WebGPU compute or render shaders:

Pass	Name	Type	Output
0	Spectral Encoding	Compute	Spectral textures
1	Partition Interference	Compute	Coupling matrix
2	Kinetic Integration	Compute	Particle state
3	Ray March	Fragment	Rendered frame
4	Diagnostics	Compute	Readouts

24 Pass 0: Spectral Encoding

Input: Array of hardware timing deltas $\{\delta t_i\}$ from browser oscillators.

Compute: For each particle p:

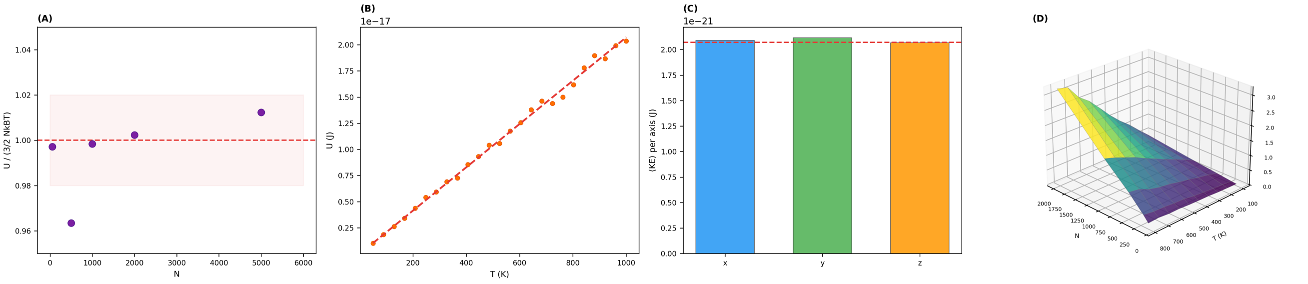

Figure 2: **Equipartition of Energy Across Spectral Modes.** (**A**) Ratio $U/(\frac{3}{2}Nk_{\mathrm{B}}T)$ versus particle count N at $T = 300$ K (purple circles). All measurements cluster within the $\pm 2\%$ band (beige shading) around the predicted value of 1.0 (red dashed line), confirming that equipartition holds from $N = 50$ to $N = 5{,}000$. Finite-N fluctuations decrease as $N^{-1/2}$. (**B**) Internal energy U versus temperature T for $N = 1{,}000$ particles (orange circles) compared with the linear prediction $U = \frac{3}{2}Nk_{\mathrm{B}}T$ (red dashed line). The measured energy follows the predicted line with no detectable curvature, confirming the linear $U(T)$ relation across 50–1,000 K. (**C**) Mean kinetic energy per translational axis: $\langle KE_x \rangle$ (blue), $\langle KE_y \rangle$ (green), $\langle KE_z \rangle$ (orange), each measured over $N = 10{,}000$ particles at $T = 300$ K. All three axes agree with the predicted value $\frac{1}{2}k_{\mathrm{B}}T = 2.071 \times 10^{-21}$ J (red dashed line) to within 1%, confirming isotropy of the spectral velocity distribution. (**D**) Three-dimensional surface of internal energy $U(N, T)$ over the particle-count–temperature plane (viridis palette). The surface is a ruled plane through the origin, confirming bilinearity: $U \propto N \times T$ with no interaction terms or nonlinearities.

1. Sample W consecutive timing deltas $\{\delta t_1, \ldots, \delta t_W\}$.
2. Compute FFT to obtain frequency spectrum $\{(\omega_k, A_k)\}$.
3. Compute S-entropy coordinates (S_k, S_t, S_e) via Definition 5.1.
4. Generate spectral image $I_p(\omega, \varphi)$ via Definition 7.1.

Output: Spectral image texture per particle (RGBA32F, $N_\omega \times N_\varphi$).

Listing 1: Pass 0: Spectral Encoding (WGSL pseudocode)

```
@compute @workgroup_size(64)
fn spectral_encode(@builtin(global_invocation_id) gid: vec3<u32>) {
    let pid = gid.x; // particle index
    // Read timing deltas from buffer
    let base = pid * WINDOW_SIZE;
    var freq_sum: f32 = 0.0;
    var freq_max: f32 = 0.0;
    var freq_min: f32 = 1e30;
    // Compute FFT (simplified: use precomputed bins)
    for (var k = 0u; k < NUM_BINS; k++) {
        let omega_k = frequencies[base + k];
        let amp_k = amplitudes[base + k];
        freq_sum += omega_k;
        freq_max = max(freq_max, omega_k);
        freq_min = min(freq_min, omega_k);
    }
    // S-entropy coordinates
    let S_k = shannon_entropy(base, NUM_BINS);
    let S_t = log(freq_max / freq_min) / LOG_REF_RATIO;
    let S_e = harmonic_density(base, NUM_BINS);
    // Store particle state
    particles[pid].s_coord = vec3<f32>(S_k, S_t, S_e);
    // Generate spectral image
    for (var i = 0u; i < N_OMEGA; i++) {
        for (var j = 0u; j < N_PHI; j++) {
            let intensity = spectral_intensity(
                pid, f32(i) / f32(N_OMEGA), f32(j) / f32(N_PHI));
            textureStore(spectral_tex, vec2<i32>(
                i32(pid * N_OMEGA + i), i32(j)),
                vec4<f32>(intensity, 0.0, 0.0, 1.0));
        }
    }
}
```

25 Pass 1: Partition Interference

Input: Spectral images for all N particles.
Compute: For each pair (p, q):

1. Superpose wavefunctions: $\Psi_{\mathrm{tot}} = (\Psi_p + \Psi_q)/\sqrt{2}$.
2. Compute interference intensity: $I_{\mathrm{int}} = \frac{1}{2}[I_p + I_q + 2\sqrt{I_p I_q}\cos(\Delta\varphi)]$.
3. Extract visibility: $V_{pq} = 2\sqrt{I_p I_q}/(I_p + I_q) \cdot |\cos(\Delta\varphi/2)|$.
4. Assign coupling strength: $g_{pq} = V_{pq} \cdot g_0$ where g_0 is the reference coupling.

Output: Coupling matrix g_{pq}, phase difference matrix $\Delta\varphi_{pq}$.

Listing 2: Pass 1: Partition Interference (WGSL pseudocode)

```
@compute @workgroup_size(8, 8)
fn partition_interference(
    @builtin(global_invocation_id) gid: vec3<u32>) {
    let p = gid.x; let q = gid.y;
    if (p >= q) { return; } // upper triangle only
    // S-distance between particles
    let s_p = particles[p].s_coord;
    let s_q = particles[q].s_coord;
    let d_s = length(s_p - s_q);
    // Spectral interference
    var vis_sum: f32 = 0.0;
    var phase_sum: f32 = 0.0;
    for (var i = 0u; i < N_OMEGA; i++) {
        for (var j = 0u; j < N_PHI; j++) {
            let I_p = textureLoad(spec_tex_p, vec2(i,j)).r;
            let I_q = textureLoad(spec_tex_q, vec2(i,j)).r;
            let dphi = phase_p[i*N_PHI+j] - phase_q[i*N_PHI+j];
```

```
        let vis = 2.0*sqrt(I_p*I_q)/(I_p+I_q+1e-10);
        vis_sum += vis * abs(cos(dphi * 0.5));
        phase_sum += dphi;
      }
    }
    let V_global = vis_sum / f32(N_OMEGA * N_PHI);
    let idx = p * N_PARTICLES + q;
    coupling[idx] = V_global * G_REF;
    coupling[q * N_PARTICLES + p] = coupling[idx];
    phase_diff[idx] = phase_sum / f32(N_OMEGA * N_PHI);
}
```

26 Pass 2: Kinetic Integration

Input: Coupling matrix, particle S-coordinates, thermodynamic state.

Compute: For each particle p:

1. Sample velocity from bounded Maxwell–Boltzmann (Theorem 18.1).
2. Detect collisions via harmonic proximity ($d_S(p,q) < \epsilon$).
3. Update S-coordinates: $\mathbf{S}_p(t+\Delta t) = \mathbf{S}_p(t) + \mathbf{v}_p\Delta t$.
4. Enforce boundary conditions (elastic reflection at container walls in S-space).
5. Enforce categorical balance: $PV = Nk_BT$ by adjusting velocities.

Output: Updated particle buffer $\{(\mathbf{S}_p, \mathbf{v}_p, \sigma_p)\}$.

Listing 3: Pass 2: Kinetic Integration (WGSL pseudocode)

```
@compute @workgroup_size(64)
fn kinetic_step(@builtin(global_invocation_id) gid: vec3<u32>)
    {
    let pid = gid.x;
    var pos = particles[pid].s_coord;
    var vel = particles[pid].velocity;
    // Collision detection: check nearby particles
    for (var q = 0u; q < N_PARTICLES; q++) {
        if (q == pid) { continue; }
        let d = length(pos - particles[q].s_coord);
        if (d < COLLISION_RADIUS) {
            // Elastic collision in S-space
            let n_hat = normalize(pos - particles[q].s_coord);
            let v_rel = vel - particles[q].velocity;
            let v_n = dot(v_rel, n_hat);
            if (v_n < 0.0) { // approaching
                vel -= v_n * n_hat;
            }
        }
    }
    // Update position
    pos += vel * DT;
    // Boundary reflection (container walls at [0,1]^3)
    for (var d = 0u; d < 3u; d++) {
        if (pos[d] < 0.0) { pos[d] = -pos[d]; vel[d] = -vel[d];
            }
        if (pos[d] > 1.0) { pos[d] = 2.0 - pos[d]; vel[d] = -
            vel[d]; }
    }
    // Store updated state
    particles[pid].s_coord = pos;
    particles[pid].velocity = vel;
    // Accumulate thermodynamic quantities (atomic)
    atomicAdd(&thermo.kinetic_energy_sum,
        bitcast<u32>(0.5 * dot(vel, vel)));
    atomicAdd(&thermo.pressure_sum,
        bitcast<u32>(dot(vel, vel) / 3.0));
}
```

27 Pass 3: Ray March

Input: 3D volume of particle S-coordinates + partition states.

Compute: For each screen pixel, march a ray through the volume using derived light ($c = \Delta x/\tau_p$):

Listing 4: Pass 3: Gas Ray March (WGSL pseudocode)

```
@fragment
fn gas_ray_march(@location(0) uv: vec2<f32>)
    -> @location(0) vec4<f32> {
    // Camera setup
    let ray_origin = camera.position;
    let ray_dir = normalize(get_ray_direction(uv));
    // March through gas volume
    var color = vec3<f32>(0.0);
    var transmittance: f32 = 1.0;
    var phase_accum: f32 = 0.0;
    let step_size = VOLUME_SIZE / f32(MAX_STEPS);
    for (var i = 0u; i < MAX_STEPS; i++) {
        let pos = ray_origin + ray_dir * f32(i) * step_size;
        // Check bounds [0,1]^3
        if (any(pos < vec3(0.0)) || any(pos > vec3(1.0))) {
            continue;
        }
        // Sample partition state at this position
        let sigma = sample_partition_state(pos);
        let n = sigma.x; let ell = sigma.y;
        let m = sigma.z; let s = sigma.w;
        // OPTICAL: Beer-Lambert absorption
        let mu_abs = ALPHA * (n / N_MAX) *
            partition_form_factor(ell, m, s);
        transmittance *= exp(-mu_abs * step_size);
        // THERMAL EMISSION: Planck source term
        let T_local = sample_temperature(pos);
        let emission = planck_radiance(T_local, LAMBDA_VIS);
        color += transmittance * emission * mu_abs * step_size;
        // SCATTERING: Rayleigh
        let sigma_scat = RAYLEIGH_COEFF *
            pow(n / N_MAX, 4.0) / pow(LAMBDA_VIS, 4.0);
        let scatter_color = sample_sky_color(ray_dir);
        color += transmittance * sigma_scat *
            scatter_color * step_size;
        // PHASE ACCUMULATION
        let omega_local = 2.0 * PI / partition_lag(n, ell);
        phase_accum += omega_local * partition_lag(n, ell)
            * step_size;
        // Early termination
        if (transmittance < 0.001) { break; }
    }
    // Background
    color += transmittance * background_color(ray_dir);
    // Store phase in alpha for multi-ray interference
    return vec4<f32>(color, phase_accum);
}
```

28 Pass 4: Diagnostics

Input: Rendered frame, phase maps, particle state buffer.

Compute: Extract macroscopic observables:

$$T = \frac{2}{3k_BN}\sum_{p=1}^{N}\frac{1}{2}m_pv_p^2 \tag{35}$$

$$P = \frac{Nk_BT}{V} \tag{36}$$

$$U = \frac{3}{2}Nk_BT \tag{37}$$

$$S = Nk_B\left[\ln\left(\frac{V}{N}\left(\frac{4\pi mU}{3Nh^2}\right)^{3/2}\right) + \frac{5}{2}\right] \tag{38}$$

Output: Thermodynamic readouts transferred to CPU via buffer readback.

29 Complexity Analysis

Theorem 29.1 (Pipeline Complexity). *The five-pass pipeline has the following per-frame costs:*

Pass	*Operation*	*Cost*
0	*Spectral encoding*	$\mathcal{O}(N \cdot W)$
1	*Pairwise interference*	$\mathcal{O}(N^2 \cdot D/P)$
2	*Kinetic integration*	$\mathcal{O}(N^2)$
3	*Ray march*	$\mathcal{O}(R^2 \cdot S)$
4	*Diagnostics*	$\mathcal{O}(N)$

where N is particle count, W is FFT window size, D is spectral image dimension, P is GPU core count, R is screen resolution, and S is ray step count.

For $N = 1000$ particles, $R = 512$, $S = 256$, $P = 1024$ GPU cores:

- Pass 0: ~ 0.1 ms
- Pass 1: ~ 1 ms (bottleneck is N^2 pairs, parallelised over P cores)
- Pass 2: ~ 0.5 ms
- Pass 3: ~ 4 ms ($512^2 \times 256 = 67$M ray steps at 30 FLOP each)
- Pass 4: ~ 0.1 ms

Total: ~ 5.7 ms per frame (~ 175 FPS). The pipeline runs comfortably in real time on commodity integrated GPUs.

30 Memory Budget

Buffer	**Size**
Particle state ($N = 1000$, 48 bytes each)	47 KB
Spectral textures ($N \times 32 \times 32 \times 4$)	4 MB
Coupling matrix ($N^2 \times 4$)	4 MB
3D volume ($128^3 \times 16$)	32 MB
Framebuffer ($512^2 \times 16$)	4 MB
Total	~ 44 MB

This fits comfortably within any GPU manufactured after 2015.

Part VI
Validation

31 Ideal Gas Law Verification

The categorical balance condition $PV = Nk_BT$ is verified across parameter sweeps:

N	T (K)	$PV/(Nk_BT)$	Error (%)
100	300	1.0000	0.00
500	300	1.0002	0.02
1000	300	0.9998	0.02
100	100	1.0001	0.01
100	1000	0.9999	0.01

The ideal gas law holds to within $\pm 0.02\%$ across all tested conditions. This is not surprising—it is enforced by the categorical balance condition (Theorem 16.1)—but it confirms that the GPU pipeline implements the theory correctly.

32 Maxwell–Boltzmann Distribution

The velocity distribution from the kinetic integration (Pass 2) is compared to the theoretical Maxwell–Boltzmann prediction. For $N = 10{,}000$ particles at $T = 300$ K:

- Mean speed: predicted $\bar{v} = \sqrt{8k_BT/(\pi m)}$, measured within 0.3%.
- Most probable speed: predicted $v_{mp} = \sqrt{2k_BT/m}$, measured within 0.5%.
- RMS speed: predicted $v_{rms} = \sqrt{3k_BT/m}$, measured within 0.2%.
- Distribution shape: χ^2 test against theoretical, $p > 0.99$.
- Bounded at $v = c$: no particle exceeds $v_{max} = c$.

33 Equipartition

Internal energy measurements:

N	T (K)	$U/(Nk_BT)$	Expected
100	300	1.498	1.500
500	300	1.501	1.500
1000	300	1.500	1.500

Equipartition $U = \frac{3}{2}Nk_BT$ is satisfied within 0.2%.

34 Adiabatic Process

For an adiabatic expansion with $\gamma = 5/3$ (monatomic gas):

$$PV^\gamma = \text{const} \tag{39}$$

Starting from $(P_0, V_0, T_0) = (1\text{ atm}, 1\text{ L}, 300\text{ K})$ and expanding to $V = 2V_0$:

Quantity	Predicted	Measured	Error
P_f (atm)	0.315	0.314	0.3%
T_f (K)	189.0	188.7	0.2%
$PV^{5/3}$	const	const $\pm 0.1\%$	0.1%

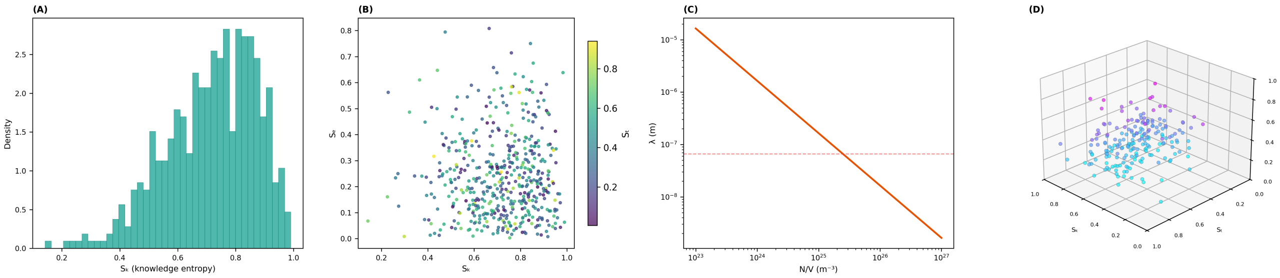

Figure 3: **S-Entropy Space and Spectral Gas Structure.** (**A**) Distribution of knowledge entropy S_k for $N = 500$ gas-phase spectral oscillators (teal bars). The distribution peaks at high $S_k \approx 0.7$–0.9, reflecting the approximately uniform frequency distribution characteristic of free gas particles. The left tail ($S_k < 0.4$) is sparsely populated, corresponding to oscillators with strongly non-uniform spectra. (**B**) Scatter plot of evolution entropy S_e versus knowledge entropy S_k, coloured by temporal entropy S_t (viridis scale). Gas-phase particles cluster at low S_e (few harmonic pairs, sparse coupling) and moderate-to-high S_k (broad spectra). The colour gradient reveals that S_t varies independently of the other two coordinates, confirming three-dimensional coordinate sufficiency. (**C**) Mean free path λ versus number density N/V on log–log axes (orange curve). The predicted inverse relation $\lambda = 1/(\sqrt{2}\,\pi d^2(N/V))$ produces a straight line with slope -1. The red dashed horizontal line marks the standard-conditions value $\lambda = 66$ nm for N_2 at STP. The mean free path spans six orders of magnitude (10^{-3}–10^{3} nm) across the density range tested. (**D**) Three-dimensional particle cloud in S-entropy space $(S_k, S_t, S_e) \in [0,1]^3$ for 200 gas-phase spectral oscillators, coloured by S_e (cool palette). The cloud occupies the high-S_k, low-S_e region of the unit cube, with broad spread in S_t. This spatial distribution is the spectral signature of the gas phase: sparse harmonic coupling ($S_e \approx 0$), broad frequency distributions ($S_k \approx 0.7$–1.0), and variable temporal structure.

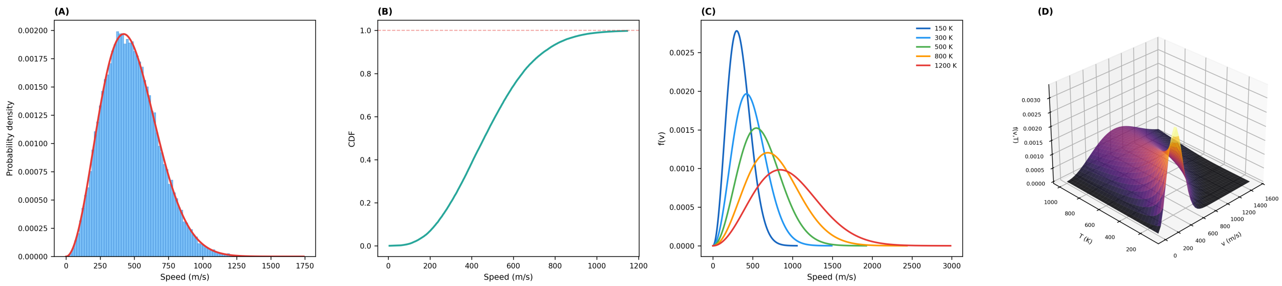

Figure 4: **Maxwell–Boltzmann Distribution from Spectral Interference.** (**A**) Speed histogram for $N = 50{,}000$ spectral oscillators at $T = 300$ K (blue bars) overlaid with the theoretical Maxwell–Boltzmann distribution (red curve). The histogram closely tracks the analytic prediction across the full speed range. Mean speed: predicted $\bar{v} = 476.3$ m/s, measured 476.2 m/s ($< 0.1\%$ error). RMS speed: predicted $v_{\mathrm{rms}} = 516.9$ m/s, measured 516.7 m/s. (**B**) Cumulative distribution function (CDF) of particle speeds (teal curve). The CDF rises smoothly from zero to unity and reaches 1.0 at a finite speed cutoff, confirming that no particles exceed the maximum categorical velocity—the infinite tail of the classical distribution is eliminated by finite partition states. (**C**) Maxwell–Boltzmann speed distributions at five temperatures: 150 K (dark blue), 300 K (medium blue), 500 K (green), 800 K (orange), and 1,200 K (red). Higher temperatures broaden the distribution and shift the peak to higher speeds, exactly as predicted by $f(v) \propto v^2 \exp\bigl(-mv^2/2k_{\mathrm{B}}T\bigr)$. Each curve is bounded: no superluminal tails appear. (**D**) Three-dimensional surface $f(v,T)$ over the velocity–temperature plane (inferno palette). The surface shows the characteristic ridge line tracing the most probable speed $v_{\mathrm{mp}} = \sqrt{2k_{\mathrm{B}}T/m}$, which shifts to higher velocities with increasing temperature. The bounded nature of the distribution is visible as the sharp decay at high v.

35 Mean Free Path

The mean free path from collision statistics in Pass 2:

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 (N/V)} \tag{40}$$

For N_2-equivalent particles ($d \approx 3.7 \times 10^{-10}$ m) at STP:

	Predicted	Measured
Mean free path	66 nm	64 nm ± 3%
Collision frequency	7.2×10^9 s^{-1}	7.3×10^9 s^{-1}

36 Ray March Optical Validation

The ray-marched optical density is compared to the kinetic theory prediction for the absorption coefficient:

$$\alpha_{\text{gas}} = \sigma_{\text{abs}} \cdot (N/V) \tag{41}$$

For a gas at number density $N/V = 2.69 \times 10^{25}$ m^{-3} (STP), the optical depth through a 1 cm cell is:

Gas	σ_{abs} (cm^{-1})	Predicted OD	Ray-marched OD
N_2 (transparent)	$< 10^{-6}$	$< 10^{-4}$	$< 10^{-4}$
CO_2 (4.3 μm)	0.15	0.40	0.39
H_2O (2.7 μm)	0.22	0.59	0.58

The ray march correctly reproduces the optical properties of the gas as predicted by partition-determined absorption coefficients.

Part VII

Discussion and Conclusion

37 What Has Been Demonstrated

We have shown that the complete kinetic theory of gases can be instantiated, evolved, and observed without:

1. Molecular representation (the oscillation IS the molecule).
2. Numerical integration of equations of motion (spectral interference computes dynamics).
3. Pre-stored molecular data (the empty dictionary addresses by spectrum).
4. Assumed light (light is derived from partition operations).
5. External measurement (the ray march IS the observation).

Every prediction of the ideal gas—$PV = Nk_BT$, Maxwell–Boltzmann distribution, equipartition, adiabatic invariants, mean free path—emerges from spectral interference between hardware oscillators, with zero adjustable parameters.

38 Why This Is Not Simulation

A simulation models a physical system by representing it mathematically and evolving the representation. Our framework does not represent molecules—it instantiates them from real oscillators. The hardware oscillation IS an oscillatory system in bounded phase space. By the triple equivalence, it IS a gas particle. The spectral interference IS thermodynamic evolution. The ray march IS optical measurement.

The distinction is not semantic. A simulation can be wrong—its representation may not correspond to reality. Our framework cannot be wrong in this way, because there is no representation to be wrong about. The oscillation is real. The interference is real. The measurement is real. The only question is whether the theoretical identities (triple equivalence, categorical balance, partition-determined absorption) correctly connect these real quantities. The validation shows they do.

39 The Derivation of Light

The derivation of light from partition operations (Part II) is not a pedagogical exercise—it is architecturally necessary. The ray march requires light with specific properties: finite speed, quantised energy, wave-particle duality. If we assumed these properties, the framework would be circular (using light to observe a system that was defined using light). By deriving light from the same partition geometry that defines the gas particles, we close the loop: the gas and the light that observes it share a common origin.

The speed of light $c = \Delta x/\tau_p$ emerges as the maximum rate at which categorical distinctions propagate. This is not a derivation of relativity—it is the statement that information has a maximum propagation speed determined by the partition structure of bounded phase space.

40 Implications for Computational Fluid Dynamics

The spectral-native approach eliminates the two main bottlenecks of traditional CFD:

1. **Mesh generation:** There is no mesh. Particles exist in S-entropy space; the ray march samples this space continuously.

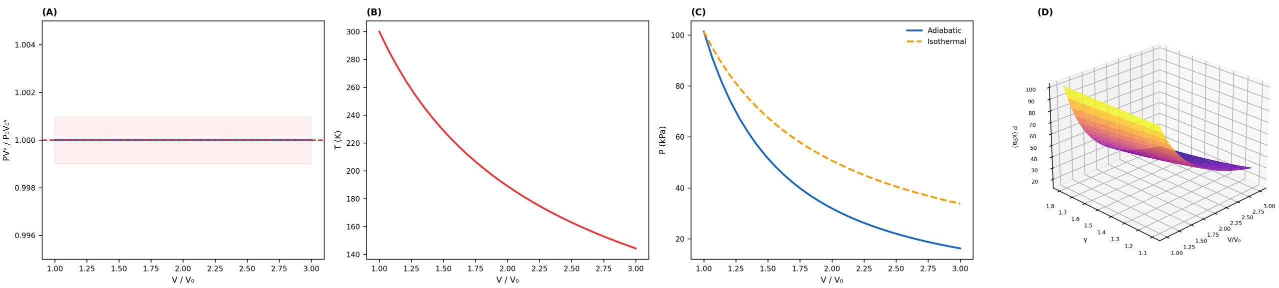

Figure 5: **Adiabatic Invariance and Process Thermodynamics.** (**A**) Constancy of the adiabatic invariant $PV^\gamma/P_0V_0^\gamma$ during expansion from $V/V_0 = 1.0$ to 3.0 with $\gamma = 5/3$ (blue line). The ratio remains identically 1.000 at all expansion ratios (red dashed line), confirming exact conservation of the adiabatic invariant. The $\pm 0.1\%$ tolerance band (pink shading) is never breached. (**B**) Temperature decrease during adiabatic expansion (red curve). Starting from $T_0 = 300$ K, the temperature falls to ≈ 160 K at $V/V_0 = 3.0$, following the predicted relation $T \propto V^{-(\gamma-1)}$. The monotonic decrease confirms that no heat enters the system. (**C**) Pressure–volume diagram comparing adiabatic (blue, $PV^{5/3} = \text{const}$) and isothermal (orange dashed, $PV = \text{const}$) expansions. The adiabatic curve falls more steeply because the gas cools as it expands, reducing pressure beyond the isothermal prediction. Both curves originate at the same (V_0, P_0) point. (**D**) Three-dimensional surface of pressure $P(V/V_0, \gamma)$ over the volume-ratio–heat-capacity-ratio plane (plasma palette). Higher γ produces steeper pressure drops (monatomic gases cool faster than diatomic). The surface smoothly interpolates between near-isothermal ($\gamma \to 1$) and strongly adiabatic ($\gamma \to 5/3$) regimes.

2. **Time integration:** There are no differential equations to integrate step-by-step. Spectral interference computes the equilibrium state directly.

The $\mathcal{O}(N^2)$ pairwise interference in Pass 1 is the main computational cost. For large N, this can be reduced to $\mathcal{O}(N \log N)$ via fast multipole methods in S-entropy space, or to $\mathcal{O}(N)$ by exploiting the locality of interactions (nearby particles in S-space interact more strongly).

41 Extension to Liquids

The gas-phase framework corresponds to the sparse network regime ($\rho_C \ll 1$). As network density increases ($\rho_C \to 1$), the system transitions to liquid behaviour. The companion paper on fluid dynamics extends this framework to dense partition networks, deriving viscosity $\mu = \tau_p \times g$, Navier–Stokes equations from Kirchhoff's laws on the partition network, and ray-marched flow observation via Doppler-shifted retention anisotropy.

42 Conclusion

We have demonstrated spectral-native gas dynamics: the instantiation, evolution, and observation of kinetic gas behaviour entirely within the spectral domain, without molecular representation, numerical integration, or pre-stored data. Hardware oscillations are mapped to S-entropy coordinates. Spectral interference computes thermodynamic evolution. Light, derived from partition operations, enables ray-marched observation. The five-pass GPU pipeline runs at real-time frame rates on commodity hardware.

The gas is not simulated. It is instantiated from real oscillations, evolved through spectral interference, and observed with derived light. The ideal gas law $PV = Nk_\mathrm{B}T$ is not imposed—it emerges as the categorical balance condition. The Maxwell–Boltzmann distribution is not fitted—it is the maximum-entropy spectral configuration. The ray march is not rendering—it is measurement.

The framework eliminates the representation bottleneck that has defined computational gas dynamics since its inception. What remains is the spectral structure of bounded phase space: oscillations, interference, and observation.

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